





## Global Sections

A fundamental result used throughout MOS is

$$H^0(C, \mathcal{F}) = \ker(L).$$

Consequently, global sections are precisely those assignments whose disagreement energy vanishes. Equivalently,

$$\omega(x) = 0 \iff x \in H^0(C, \mathcal{F}).$$

This provides a direct computational characterization of coherent memory states.

## Normalized Discord

For comparisons across different systems and scales, MOS introduces the dimensionless normalized discord

$$\rho = \frac{\omega(x)}{|B[x]^2|}, \quad 0 \leq \rho \leq 1,$$

where

$$B$$

is an appropriate normalization constant. The normalized quantity removes dependence on the overall scale of the assignment, allowing coherence values to be compared across distinct memory structures.



The coherence functional is one of the central diagnostic tools of MOS. It transforms the abstract notion of compatibility into a computable scalar quantity while preserving its geometric interpretation through the sheaf Laplacian. Subsequent sections build upon this measure to develop semantic geometry, diagnostics and adaptive structural dynamics.

## 6. Semantic Geometry

The combinatorial structure of the simplicial complex and the local constraints imposed by the cellular sheaf describe how information is organized. Semantic geometry introduces a complementary question:

[How should concepts themselves be represented?]

Within MOS, every concept is modeled probabilistically as an uncertainty distribution rather than a single point in space. This representation preserves both the estimated semantic location and the associated confidence of that estimate.

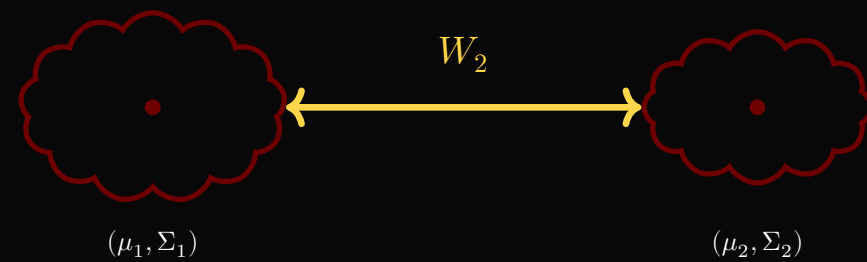
$$[X_i \sim \mathcal{N}(\mu_i, \Sigma_i)]$$

Each concept is represented by

$$(\mu_i, \Sigma_i),$$

where

- $\mu_i$  denotes the semantic embedding,
- $\Sigma_i$  represents uncertainty,
- higher confidence corresponds to smaller covariance.



## Semantic Distance

The similarity between two concepts is measured geometrically rather than combinatorially. MOS adopts the Wasserstein metric to compare probabilistic semantic representations.

$$[W_2(X_1, X_2)]$$

The resulting distance reflects differences in both semantic position and uncertainty. Unlike Euclidean distance alone, this comparison incorporates the full probabilistic structure of the concepts.

## Precision-Weighted Fusion

When multiple compatible concepts are combined, MOS performs a precision-weighted fusion. Concepts with greater certainty contribute proportionally more to the resulting estimate. The fused representation is obtained through precision averaging.

$$\mu = \left( \sum_i \Sigma_i^{-1} \right)^{-1} \left( \sum_i \Sigma_i^{-1} \mu_i \right)$$

The associated covariance decreases as mutually consistent evidence is combined.



## Interpretation

Semantic geometry complements the topological organization of MOS.

- Topology  $\implies$  Which concepts are related?
- Geometry  $\implies$  How similar are they?
- Probability  $\implies$  How certain is that relationship?

Together, these components provide a mathematical description of both the structure and the uncertainty of conceptual memory.

Semantic geometry does not replace the simplicial complex or the cellular sheaf. Instead, it enriches the existing topological framework by introducing a continuous geometric layer capable of measuring similarity, uncertainty and information fusion. The topological and geometric components therefore play complementary roles within the overall MOS architecture.

## 7. Topological Diagnostics

Computing a coherence score answers only one question:

[Is the memory coherent?]

When disagreement is detected, a second problem immediately arises:

[Where does the inconsistency originate?]

MOS addresses this through topological diagnostics. Rather than treating inconsistency as a single scalar quantity, the framework identifies the simplices responsible for incompatible local assignments and analyzes their surrounding topological structure.

## Topological Invariants

Once problematic regions have been identified, global topological properties provide additional structural information. Representative invariants include

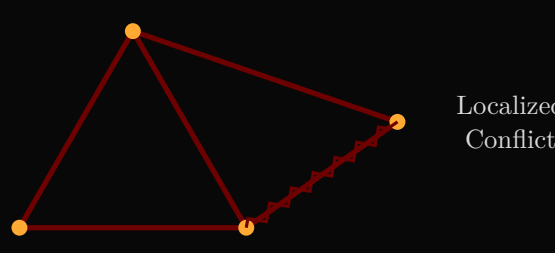
$$[b_0]$$

(number of connected components)

and

$$[H^1(C, \mathcal{F})]$$

which detects non-trivial compatibility obstructions within the sheaf. These quantities distinguish isolated local disagreement from genuine global structural inconsistency.



## Interpretation

Diagnostics transform coherence from a passive measurement into an active verification procedure.

[Measure  $\rightarrow$  Locate  $\rightarrow$  Analyze  $\rightarrow$  Repair]

The objective is not only to detect inconsistency but also to determine its geometric origin within the underlying simplicial structure.

The diagnostic procedures presented here are intended as mathematical tools for structural verification. Their implementation within a complete cognitive architecture remains future work. The present framework establishes the mathematical quantities and topological invariants upon which such procedures can be constructed.

## 9. Dynamics: Growth Operators and the Operator Algebra

Unlike a static database, the Mathematical Operating System evolves by modifying its underlying mathematical structure. Learning is interpreted as a change in shape: concepts are reorganized, functional coalitions strengthen or decay, sheaf data adapts, and successful procedures become permanent mathematical operators. The evolution of MOS is therefore governed not only by changes in memory, but also by changes in the collection of admissible operations.

$$[\mathcal{D}(t) \longrightarrow \mathcal{D}(t+1)]$$

The operator algebra is itself dynamic. Learning expands not only knowledge but also the mathematical operations available for future reasoning.

## The Four Growth Axes

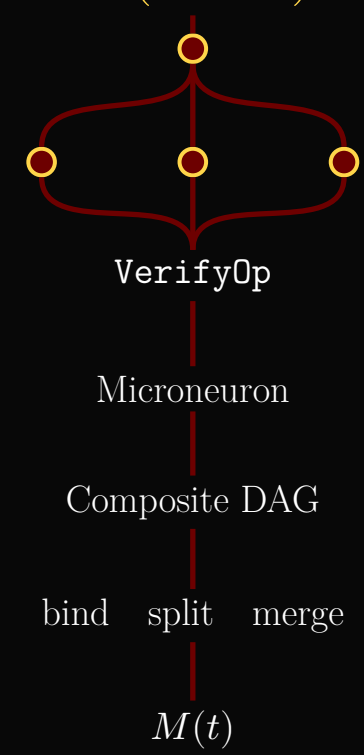
Evolution proceeds simultaneously along four independent directions.

$$[\delta C_{\text{fine}}, \quad \delta C_{\text{coarse}}, \quad \delta \mathcal{F}, \quad \delta \mathcal{D}]$$

- $\delta C_{\text{fine}}$  Growth of concepts (semantic memory).
- $\delta C_{\text{coarse}}$  Formation and restructuring of coalitions (functional connectivity).
- $\delta \mathcal{F}$  Evolution of stalks and restriction maps within the cellular sheaf.
- $\delta \mathcal{D}$  Growth of the operator algebra through the acquisition of new procedural skills.

The fourth axis distinguishes MOS from a fixed mathematical model: the collection of admissible operators evolves over time.

$$\mathcal{D}(t+1)$$



Learning as Structural Growth

## Structural Operators

The elementary structural moves acting on the simplicial complex are

[bind, collapse, split, merge].

Every operation is recorded in an immutable mutation history.

### Remark.

The earlier terminology *Pachner move* was intentionally discarded. A knowledge complex is not a manifold, therefore Pachner invariance does not apply.

## Hebbian Coupling and Structural Decay

Each coalition

$$\sigma$$

is assigned a weight

$$w(\sigma, t) \in [0, 1].$$

Repeated co-activation strengthens the coalition

[Fire Together  $\implies$  Wire Together.]

Idle coalitions decay according to

$$w(\sigma, t+1) = (1 - \gamma) w(\sigma, t).$$

Coalitions whose weights fall below the binding threshold collapse, providing an intrinsic form of structural garbage collection. Pairs whose weights exceed the threshold become 1-simplices (edges) of the knowledge complex.

## Persistence and Microneurons

A weight filtration is analyzed using persistent homology. Coalitions and concept-cycles that persist across the filtration are promoted into permanent mathematical structure. A particularly important promoted object is the **microneuron**:

- an operadic sub-tree,
- represented internally as a composite computational DAG,
- repeatedly resolves conflict,
- repeatedly passes verification,
- promoted into a new atomic generator of  $\mathcal{D}$ .

The promotion gate is

[VerifyOp]

Only externally verified composite procedures become permanent generators of the operator algebra.

## Mathematical Interpretation

MOS evolves simultaneously on four interconnected levels:

[Concepts  $\rightarrow$  Coalitions  $\rightarrow$  Sheaf Data  $\rightarrow$  Operator Algebra.]

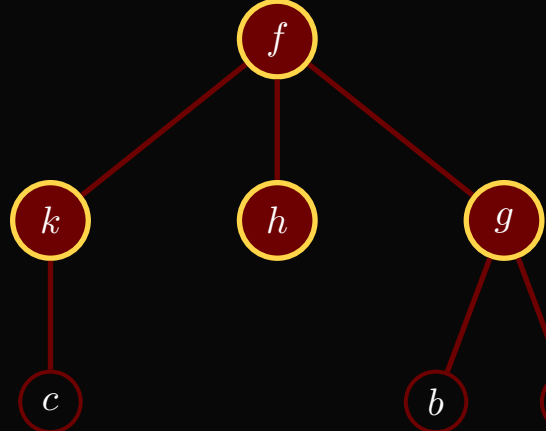
Unlike the preceding three growth mechanisms, the final axis changes the very collection of admissible mathematical operations, allowing the system to acquire new procedural primitives over time.

## 10. Operadic Computation and Parallel Scheduling

Within MOS, a reasoning act is represented as a directed acyclic graph (DAG) of operator nodes rather than a linear sequence of computations. Each node corresponds to an operator, while directed edges encode dependency relations. Independent computations therefore become explicit, allowing reasoning to be executed in parallel whenever dependencies permit.

$$T = (N, \prec)$$

A reasoning program is a partially ordered set of operator nodes whose execution order is determined by the dependency relation  $\prec$ .



## Operadic Foliation

The scheduler partitions the reasoning DAG into successive *antichain layers*.

$$L_0, L_1, L_2, \dots$$

where

$$L_0 = \{\text{minimal nodes}\}$$

and

$$L_{t+1} = \{n : \text{all predecessors belong to } L_{t'}\}.$$

Each layer is executed in parallel because no two nodes within the same antichain possess a dependency relation.

## Support Sets

Each operator records the subset of fine-complex vertices that it reads and writes. Operators with disjoint support sets may safely execute concurrently, providing a mathematically justified scheduling rule rather than a heuristic approximation.

## Interpretation

Reasoning is viewed as composition rather than sequential execution. The partial order determines correctness, while the foliation determines parallel execution.

## 11. The Lift-Project Adjunction

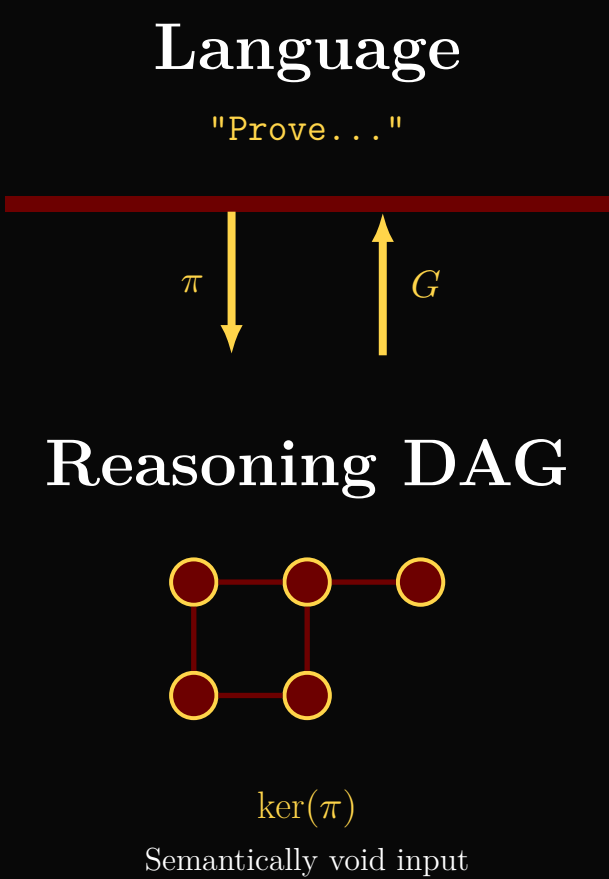
MOS distinguishes two complementary representations of reasoning. Natural language serves as the interface through which problems are posed, while internally the system operates on validated reasoning DAGs. Two boundary morphisms connect these worlds:

$$[\pi : \text{Language} \longrightarrow \text{Reasoning DAG}]$$

and

$$[G : \text{Reasoning DAG} \longrightarrow \text{Language}].$$

The lift converts a natural-language request into a validated reasoning program, while the projection renders a completed reasoning graph back into human-readable form.



## Boundary Validation

The lift operator is guarded by a validator before a reasoning program enters the computational engine. Malformed reasoning DAGs are rejected if they contain

- duplicate node identifiers,
- dangling child references,
- directed cycles,
- disconnected components,
- no terminal output node.

Only validated reasoning programs are admitted to downstream execution.

## Kernel Behaviour

Empirically, semantically irrelevant content is mapped into

$$[\ker(\pi)].$$

For example, non-reasoning text embedded inside a mathematical request generates no reasoning operators and therefore contributes nothing to the internal reasoning DAG. This observed behaviour motivates the interpretation of semantically void input as belonging to the kernel of the lift.

## Claims and Non-Claims

The notation

$$L \dashv M$$

is presented as the intended categorical organisation of the framework. The corresponding unit and counit identities have *not* been formally established in the present work. Accordingly, the adjunction should be interpreted as a guiding mathematical principle together with the concrete morphisms

$$(\pi, G),$$

rather than as a proved categorical theorem.

## 12. Verification: The External Oracle

Internal coherence and external truth are distinct concepts. The sheaf coherence measure determines whether the system should continue exploring or reconcile conflicting information, but it does not certify mathematical correctness. Truth is determined exclusively by an external symbolic verification oracle.

$$[\text{Coherence} \neq \text{Truth}]$$

Verification therefore acts as the final gate before new mathematical knowledge is permanently incorporated into the system.

## VerifyOp

Every candidate mathematical result is submitted to an external, sandboxed symbolic oracle before promotion into permanent memory.

## Three-Valued Verification Logic

The verification oracle returns one of three possible outcomes.

| Exit Code | Verdict      | Effect                                                                                                                          |
|-----------|--------------|---------------------------------------------------------------------------------------------------------------------------------|
| 0         | Verified     | Obstruction is cleared. The current state may crystallize into permanent structure.                                             |
| 1         | Refuted      | The obstruction counter is increased<br>$\Omega \leftarrow \Omega + 1000$ ,<br>forcing the controller into <b>Resolve</b> mode. |
| 2         | Unverifiable | No change is made to the obstruction counter. The computation remains unresolved.                                               |

## VerifyOp



## Absence of Proof Is Not Proof of Absence

Earlier versions of MOS treated every nonzero exit code as a failure. This incorrectly identified

$$[\text{Refuted} \equiv \text{Unverifiable}],$$

causing unresolved computations and oracle failures to be interpreted as mathematical contradictions. The present framework explicitly separates these outcomes. An oracle crash likewise returns **Unverifiable**.

## Connection to Dynamic Growth

The external oracle provides the promotion gate introduced in Construction 9.4. Only composite procedures that successfully pass

[VerifyOp]

may become permanent generators of the evolving operator algebra

$$\mathcal{D}.$$

Consequently, procedural growth is controlled by external verification rather than internal confidence alone.

## Central Principle

MOS separates three fundamentally different notions:

$$[\text{Coherence} \neq \text{Verification} \neq \text{Truth}].$$

Coherence measures internal compatibility. Verification evaluates mathematical correctness. Truth is never inferred solely from structural consistency.

## 13. Coherence-Gated Controller

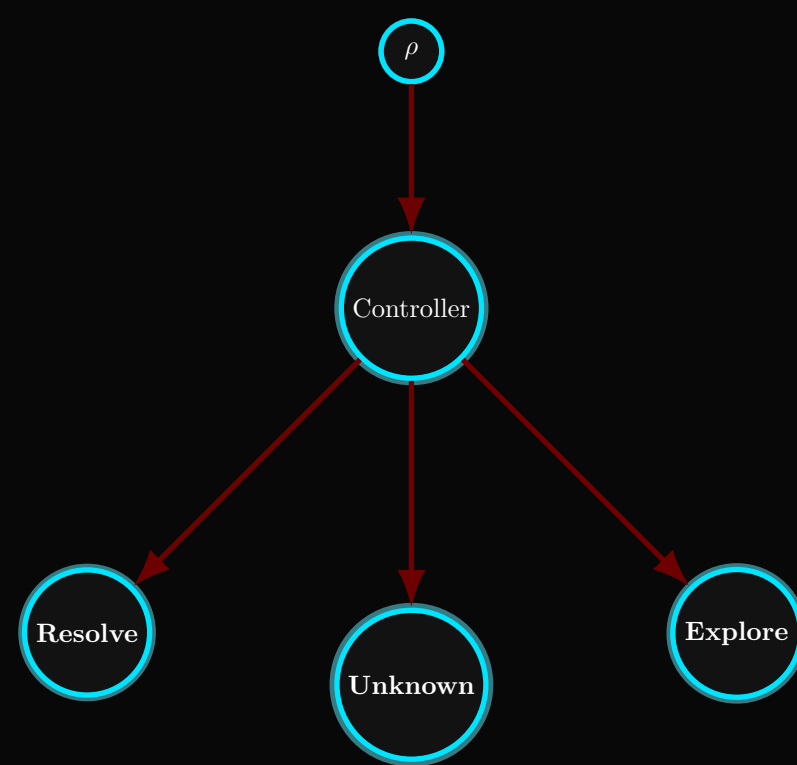
The controller determines the system's operating mode using the normalized coherence measure

$$\rho.$$

Rather than continuously expanding the knowledge complex, MOS first asks whether the current mathematical state is sufficiently coherent to justify further exploration. This decision is governed by the coherence threshold

$$\varepsilon.$$

$$\text{mode}(x) = \begin{cases} \text{Resolve,} & \rho > \varepsilon, \\ \text{Explore,} & \rho \leq \varepsilon, \\ \text{Unknown,} & \rho = \perp. \end{cases}$$



## Controller Modes

**Resolve**  
High discord. Reconcile conflicting structures before further expansion.  
**Explore**  
Coherence remains acceptable. Additional reasoning and structural growth may proceed.  
**Unknown**  
No coherence measurement is available. Importantly,

$$[\text{Unknown} \neq \text{Explore}].$$

Absence of measurement is not interpreted as evidence of consistency.

## Empirical Threshold

The manuscript reports representative coherence values obtained from 384-dimensional embeddings:

$$\begin{aligned} \text{Agreement} &\approx 0.00, \\ \text{Related Concepts} &\approx 0.06, \\ \text{Off-Topic} &\approx 0.14. \end{aligned}$$

These observations motivate

$$[\varepsilon \approx 0.10]$$

as an initial operating threshold. Two cautions remain important.

- The normalized discord is not necessarily monotonic as the number of active organs increases.
- Comparisons across differently-sized complexes require additional care.

## 14. Claims, Non-Claims and Future Directions

MOS should be interpreted as a mathematical framework for structured knowledge organization rather than a completed intelligent system. The present work establishes the mathematical architecture and introduces tools for measuring coherence, structural growth and verification. Several directions identified in the manuscript remain future work.



## Discord Localization

The coherence functional identifies the presence of inconsistency. Local diagnostic procedures determine the simplices whose restriction maps contribute most strongly to the disagreement energy. These regions become natural candidates for

- refinement,
- correction,
- semantic revision,
- structural expansion.

Rather than rebuilding the entire knowledge structure, MOS attempts to localize disagreement before modifying the underlying complex.



Toward Constitutive Reasoning

The manuscript identifies four developments that would move the sheaf from a diagnostic role toward a constitutive computational role.

- 1. Sheaf diffusion as inference

$$\dot{x} = -Lx$$

converging toward the least-discord global section.

- 2. Non-trivial learned restriction maps.
- 3. Higher-cell Laplacians for higher-order constraint propagation.
- 4. Persistence-driven structural learning through the growth mechanism introduced in Section 9.

Creativity Boundary

Following the discussion in the manuscript, MOS presently supports combinatorial and exploratory creativity through structured recombination within a fixed representational space. Transformational creativity remains limited. One genuine avenue for expansion is the growth operator

$$\delta\mathcal{D}_i$$

through which verified procedures become new primitive operators, expanding the effective action space of the system.

What is claimed?

- ✓ measurable coherence
- ✓ structured memory
- ✓ verifiable growth
- ✓ mathematical substrate

What is NOT claimed?

- ✗ intelligence proven
- ✗ AGI
- ✗ universal reasoning